

William Rochira

Systems Engineer - Scientific Computing - AI/ML

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Professional Summary

Cross-disciplinary engineer with expertise spanning performance-focused software, embedded hardware, and machine learning. PhD-qualified researcher in computational biophysics and molecular simulation. Specialised in architecting distributed systems and microservices for high-performance computing. Proven commercial delivery of end-to-end AI/ML systems, from data pipelines through to deployment and monitoring. Strong software engineering discipline in Python and C, alongside hardware expertise with microcontrollers and FPGAs. Active in generative modelling and LLMs, with a track record of converting research into robust tooling. High ownership and independent delivery, evidenced by a substantial portfolio of multi-year, self-led projects. Adaptable, quick to learn, and effective in collaborative environments.

Technical Skills

Languages: [Python](#) • Embedded C • SystemVerilog • SQL • JavaScript/TypeScript

AI & Scientific: [PyTorch](#) • [Scikit-learn](#) • JAX • RAG • RDKit • GROMACS • Qiskit

Systems & Architecture: [Distributed Systems](#) • [Microservices](#) • API Design • Vector Databases

Embedded & Hardware: [FPGA \(Vivado\)](#) • [KiCad](#) • Arduino / RP2040 • HIL Testing • SPI/I2C/PWM

Engineering & DevOps: [Docker](#) • Kubernetes • AWS • CI/CD • Linux/CLI • Git • TDD

Professional Experience

Graduate Researcher (PhD) @ [University of Oxford](#) (Oxford, UK) 2021 - 2025

- Conceived and delivered an end-to-end research project for a novel high-performance computing architecture, taking it from first-principles concept to validated hardware/software prototypes.
- Delivered a comprehensive research programme spanning de novo simulation software, physics-informed neural networks, and hardware implementations using analog electronics, microcontrollers, and FPGAs.
- Developed real-time embedded firmware for RP2040-based control planes, managing deterministic scheduling and high-speed ADC/DAC interfacing.
- Implemented hybrid quantum-classical constrained-optimisation workflow for cyclic peptide conformational search, applying VQE/QAOA with a molecular mechanics objectives and constraint handling.

Technical Business Analyst @ [Acturis](#) (London, UK) 2020 - 2021

- Owned the full-lifecycle delivery of a production machine-learning product, transforming large-scale insurance data into real-time renewal predictions.
- Architected SQL and Python pipelines with quality gates, ensuring integrity across multi-million record datasets.
- Developed models with time-aware validation, SHAP explanations, and domain-informed feature engineering.
- Productionised the full ML lifecycle via Docker and REST APIs.

Web Developer @ [TrainTaxi](#) (Peterborough, UK) 2019 - 2020

- Took full ownership of a live business-critical platform, managing the full software stack and operations.
- Re-engineered a legacy Rails and Django stack with MySQL, modernising workflows and system integrations.
- Improved database performance through indexing and query refactoring.
- Hardened platform security via least-privilege controls, patching, and robust input validation.

Selected Technical Projects

Machine Learning for Molecular Dynamics Acceleration

- Developed energy-stable neural surrogates to replace expensive analytic kernels in high-fidelity simulations.
- Built physics-informed, symmetry-aware architectures resilient to aggressive quantisation.
- Engineered low-latency, high-throughput inference via quantisation-aware training.
- Validated long-horizon stability via comprehensive MD rollouts, ensuring zero energy-drift in production.
- **Skills:** Python, C, JAX, NumPy, PINNs, SystemVerilog, RTL, Vivado, Model Evaluation, HW/SW Co-design

Citations-First Technical Research Assistant

- Developed an integrated local RAG ecosystem with verifiable citations and hallucination-resistant design.
- Engineered a hybrid retrieval engine using vector and keyword search with Postgres pgvector.
- Fine-tuned open-source models via LoRA and DPO to enforce citation compliance.
- Architected a rigorous evaluation harness for transforming RAG performance into measurable metrics.
- **Skills:** Python, LangChain, RAG, LLMs, Vector DBs, Fine-Tuning (SFT/DPO), Model Evaluation, API Design

Multi-Agent Imperfect-Information Decision-Making Platform

- Architected a modular framework for solving complex, high-dimensional imperfect-information games.
- Developed a high-throughput C/Python Monte Carlo engine with advanced variance reduction.
- Implemented multi-agent optimisation using GNN representations and opponent modelling.
- Hardware-accelerated simulation hot-paths via custom SystemVerilog kernels for sub-microsecond scoring.
- **Skills:** Python, C, SystemVerilog, Distributed Systems, GNNs, CFR, Monte Carlo, HW Acceleration

Iris: Structural Validation Platform

- Delivered Iris: an interactive end-to-end structural model validation and reporting platform.
- Architected a unified validation engine, collapsing legacy tools into a high-performance pipeline.
- Engineered an extensible framework with a multi-threaded backend and interactive, web-based reporting.
- Shipped as the default validation standard within CCP4, used by structural biologists worldwide.
- Published as peer-reviewed research in Protein Science (2020).
- **Skills:** Python, JavaScript, C++, HTML5/SVG, UI/UX, REST APIs

Education

PhD Computational Biophysics @ University of Oxford (Oxford, UK)

2021 - 2025

- A scalable dataflow architecture for high-performance molecular dynamics
- Development of quantum computing algorithms to explore cyclic peptides

MSc Computational Chemistry @ University of York (York, UK)

2019 - 2020

- Iris Validation: computational validation of macromolecular structural models

BSc Biochemistry @ University of York (York, UK)

2016 - 2019

- Coarse-grained Monte Carlo simulations to probe confined protein diffusion
- Computational analysis of mechanical protein unfolding using atomic force microscopy